

Zero-Trust Multi-Tenant Orchestration for Distributed LLM Workloads: A High-Performance DevSecOps Framework Using RDMA and Hardware Enclaves

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DOI Verification: <https://doi.org/10.5281/zenodo.22754169>
September 2026

Abstract—As Large Language Model (LLM) training and inference transition to multi-tenant cloud environments, ensuring cryptographically isolated workload environments without sacrificing network throughput remains a paramount challenge. Traditional encryption-in-transit mechanisms introduce substantial CPU overhead, severely degrading inter-node communication speeds. In this paper, we propose a novel high-performance DevSecOps framework that integrates Remote Direct Memory Access (RDMA) over Converged Ethernet (RoCEv2) with hardware-enforced Confidential Virtual Machines (CVMs). By establishing zero-trust orchestration policies within continuous integration and continuous deployment (CI/CD) pipelines, our architecture offloads security handshakes and memory access isolation directly to hardware enclaves and SmartNICs. Experimental evaluation indicates that our proposed framework reduces gradient synchronization latency by up to 41% compared to standard TLS-over-TCP pipelines, while maintaining strict memory and network isolation boundaries across multi-tenant environments.

Index Terms—Confidential Computing, RDMA, DevSecOps, Large Language Models, Multi-Tenant Cloud Security, Distributed Inference.

1 INTRODUCTION

THE scaling laws of modern generative AI demand massive parallelization across heterogeneous compute clusters [1]. In multi-tenant enterprise environments, distributed LLM workloads frequently exchange unencrypted tensor gradients and internal activation weights across shared physical infrastructure. Standard transport-layer encryption protocol stacks (e.g., TLS-over-TCP) introduce significant CPU serialization penalties that severely throttle hardware utilization [2].

To address these security and performance bottlenecks, we introduce an integrated DevSecOps framework designed for low-latency, high-security distributed compute tasks. By leveraging hardware-backed Trusted Execution Environments (TEEs) alongside zero-copy RDMA acceleration, the architecture enforces cryptographically validated zero-trust boundaries without incurring execution performance degradation.

2 SYSTEM ARCHITECTURE

The core architecture consists of three tightly integrated infrastructure layers:

- 1) **Hardware Isolation Layer:** Execution of model parameters and attention KV-caches inside hardware-enforced Confidential Virtual Machines (CVMs) utilizing AMD SEV-SNP and Intel SGX/TDX memory encryption engines [3].
- 2) **RDMA Acceleration Layer:** Microsecond-level zero-copy memory transport bypassing the host kernel network stack through RoCEv2 capabilities offloaded directly to SmartNIC hardware [4].
- 3) **DevSecOps Pipeline Engine:** Automated remote attestation, continuous vulnerability scanning, and dynamic key rotation embedded within Kubernetes-native CI/CD orchestration workflows.

3 PERFORMANCE EVALUATION

We evaluated the proposed framework against baseline cloud deployment strategies across simulated multi-tenant cluster nodes running distributed Llama-based architecture workloads.

TABLE 1
Comparative Performance Benchmarks across Inter-Node Transport Pipelines

Pipeline Architecture	Latency (ms) per Token	Throughput (Gbps)	Zero-Trust Boundary
Standard TCP / Plaintext	42.5	94.2	None
TLS-over-TCP Pipeline	68.4	41.8	Software Only
IPSec / Host Enclave	55.1	58.6	OS-Dependent
Proposed Framework (Ours)	40.3	91.5	Hardware Enclave

As detailed in Table 1, offloading memory access authentication to the hardware boundary and utilizing zero-copy RoCEv2 transfers reduces inter-node gradient synchronization latency by approximately 41% compared to standard software TLS-over-TCP pipelines (from 68.4 ms down to

40.3 ms). Furthermore, hardware memory encryption adds minimal overhead relative to unencrypted baseline transmission while maintaining end-to-end data confidentiality.

4 CONCLUSION

This study proves that robust cryptographic security and high-performance computing are not mutually exclusive in distributed LLM infrastructure. By combining hardware-accelerated RDMA with zero-trust DevSecOps orchestration, multi-tenant AI deployments can achieve near-native execution efficiency alongside strict enterprise-grade confidentiality guarantees.

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